

OMKAR PUJAR

Robotics & Automation Engineer

+91 8310704647 | omkarpujar02@gmail.com | github.com/OP2004 | linkedin.com/in/omkar-pujar-257374237 | op2004.github.io/omkarpujar.github.io

PROFESSIONAL SUMMARY

Robotics and Automation Engineer with hands-on experience in embedded systems, CAD design, and automation, plus applied experience building AI/ML and NLP-powered software. Skilled in ROS, Arduino, Raspberry Pi, and Fusion 360, alongside machine learning and LLM-based application development using Python, LangChain, and Streamlit. Proven ability to design, prototype, and deploy systems spanning hardware automation to intelligent software tools.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

AI/ML & Generative AI: Machine Learning, NLP, Generative AI, Large Language Models (LLMs), RAG, Prompt Engineering, LangChain, Vector Search, Semantic Search, Model Evaluation

Data & Development: Pandas, NumPy, Scikit-learn, Data Preprocessing, Feature Engineering, Git, GitHub

Computer Vision: OpenCV, Image Processing, Object Detection

Robotics & Automation: ROS, Arduino, Raspberry Pi, Embedded Systems, IoT, Sensors, Fusion 360, CAD, Robotics Simulation, Automation

Frameworks & Tools: Streamlit, React, HTML, CSS, VS Code

PROFESSIONAL EXPERIENCE

AI Intern | *GlowLogics*

6 Months

- Built and deployed AI applications spanning fraud detection, dynamic pricing, demand forecasting, and document Q&A, using Python and Scikit-learn.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline with LangChain to enable semantic search and question-answering over unstructured PDF documents.
- Built and trained classification models with Scikit-learn for real-time fraud detection, improving flagging consistency across transaction datasets.
- Performed data preprocessing, feature engineering, and model evaluation to iteratively improve prediction accuracy across projects.
- Designed interactive Streamlit interfaces to expose ML models to end users, streamlining AI deployment workflows.
- Collaborated using Git and GitHub for version control, code review, and iterative development within a team workflow.

PROJECTS

Ask My PDF Bot — RAG-Based Document Chatbot | *Python, LangChain, Streamlit* | github.com/OP2004

- Engineered a Retrieval-Augmented Generation chatbot that answers natural-language questions from uploaded PDFs using semantic search and vector embeddings.
- Implemented vector-based document retrieval to surface relevant passages instead of full-document scanning, reducing manual search effort.
- Deployed via Streamlit for real-time, interactive Q&A over user-uploaded documents.

Bank Fraud Detection System | *Python, Scikit-learn, Streamlit*

- Built a supervised machine learning pipeline to classify transactions as fraudulent or legitimate using engineered behavioral and transactional features.
- Designed a real-time analytics dashboard to visualize flagged transactions and model confidence scores for faster review.
- Automated model evaluation to compare precision/recall across algorithms and select the best-performing classifier.

AI Resume Analyzer (ATS Scoring Tool) | *Python, NLP, Streamlit*

- Developed an NLP-powered ATS resume analysis tool that extracts keywords, matches skills against job descriptions, and generates a resume score.
- Automated parsing of unstructured resume text into structured skill and experience data to support faster candidate screening.

Supply Chain Demand Forecasting | *Python, Machine Learning, Streamlit*

- Built a time-series forecasting system to predict product demand trends, supporting inventory and supply-chain planning decisions.
- Engineered features from historical sales data to improve forecast reliability across multiple product categories.

Dynamic Pricing Engine | *Python, Machine Learning*

- Designed an AI-based pricing model that adjusts product prices in real time based on demand signals and market trends.

Robotics & Embedded Systems Projects | *ROS, Arduino, Raspberry Pi, Fusion 360, CAD*

- Designed a hybrid quadruped robot combining wheeled and legged locomotion for improved terrain adaptability, modeled end-to-end in Fusion 360.
- Built a Cartesian coordinate robot for precision-based industrial automation and controlled linear-axis movement.

EDUCATION

B.E., Robotics and Automation Engineering — Visvesvaraya Technological University

2020 – 2026

Pre-University Course (PCMB) — Govindram Seksaria Science PU College

2020 – 2022